AICS Practical No. 03

Dataset Splitting using Train-Test Split in Python:

Title:

Study and Implementation of Dataset Splitting into Training and Testing Sets using Scikit-Learn

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Aim:

To study and implement the **Train-Test Split** technique using Python and Scikit-Learn library for dividing a dataset into training and testing datasets for machine learning applications.

Objective:

1. To understand the importance of dataset splitting.
2. To learn how to use the train_test_split() function.
3. To divide data into training and testing datasets.
4. To prepare data for machine learning model development.

Theory:

Machine Learning models learn patterns from historical data and make predictions on unseen data.

Before training a machine learning model, the dataset must be divided into two parts:

1. Training Dataset

- Used to train the model.
- Helps the model learn patterns and relationships.
- Usually contains 70%–80% of the total data.

2. Testing Dataset

- Used to evaluate the model's performance.

- Contains unseen data.
- Usually contains 20%–30% of the total data.

Dataset splitting helps avoid **overfitting**, where the model performs well on training data but poorly on new data.

The **train_test_split()** function provided by the Scikit-Learn library is commonly used for this purpose.

Software Requirements:

Software

- Python 3.x
- Jupyter Notebook / Google Colab / VS Code

Libraries

- NumPy
- Scikit-Learn

Algorithm:

Step 1: Import the required libraries.

Step 2: Create the feature dataset (X).

Step 3: Create the target labels (y).

Step 4: Use train_test_split() to divide the dataset.

Step 5: Display the training and testing datasets.

Step 6: Analyze the output.

Program

```
import numpy as np
from sklearn.model_selection import train_test_split

X = np.array([[25,50000],[30,60000],[35,70000],
          [40,80000],[45,90000],[50,100000]])

y = np.array([0,1,0,1,0,1])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

print(X_train)
print(X_test)
print(y_train)
print(y_test)
```

Program Explanation

Import NumPy Library

```
import numpy as np
```

NumPy is used to create and manipulate arrays.

Import train_test_split Function

```
from sklearn.model_selection import train_test_split
```

Imports the function used to split data into training and testing sets.

Create Feature Dataset

```
X = np.array([[25,50000],[30,60000],[35,70000],
          [40,80000],[45,90000],[50,100000]])
```

Creates a dataset containing:

Age	Salary
25	50000
30	60000
35	70000
40	80000
45	90000
50	100000

Create Target Labels:

y = np.array([0,1,0,1,0,1])

Creates output labels.

Age	Salary	Label
25	50000	0
30	60000	1
35	70000	0
40	80000	1
45	90000	0
50	100000	1

Split Dataset

```
train_test_split(
    X, y,
    test_size=0.3,
```

```
    random_state=42
)
```

test_size = 0.3

30% data is used for testing.

random_state = 42

Ensures the same random split every time the program runs.

Print Output

```
print(X_train)
print(X_test)
print(y_train)
print(y_test)
```

Displays training and testing datasets.

Expected Output:

X_train

```
[[ 50 100000]
 [ 35  70000]
 [ 45  90000]
 [ 40  80000]]
```

X_test

```
[[25 50000]
 [30 60000]]
```

y_train

```
[1 0 0 1]
```

y_test

[0 1]

(Output may vary slightly if random_state is changed.)

Observation:

Dataset Type	Number of Records
Total Dataset	6
Training Dataset	4
Testing Dataset	2

Training data contains approximately 70% of records, while testing data contains approximately 30% of records.

Advantages:

1. Easy implementation.
2. Helps evaluate model performance.
3. Prevents overfitting.
4. Widely used in machine learning projects.
5. Improves model reliability.

Applications:

1. Machine Learning Model Training
2. Data Science Projects
3. Artificial Intelligence Systems
4. Predictive Analytics
5. Classification and Regression Problems

Result:

The dataset was successfully divided into training and testing datasets using the train_test_split() function from the Scikit-Learn library. The training dataset contained 70% of the records, while the testing dataset contained 30% of the records.

Conclusion:

Thus, the program for **splitting a dataset into training and testing sets using the Scikit-Learn train_test_split() function** was successfully implemented and executed. The obtained output confirms that the dataset was correctly divided into training and testing subsets, which can be further used for developing and evaluating machine learning models.

Viva Questions (Most Expected)

Q1. What is dataset splitting?
Dividing data into training and testing sets.

Q2. Why do we use train_test_split()?
To divide data for training and testing.

Q3. What is training data?
Data used to train the model.

Q4. What is testing data?
Data used to evaluate the model.

Q5. What is random_state?
A parameter used to generate the same random output every time.

Q6. What is test_size?
The percentage of data used for testing.

Q7. Which library provides train_test_split()?
Scikit-Learn.

Q8. What is NumPy?
A Python library used for numerical computations and arrays.

Q9. What is X?
Input features.

Q10. What is y?
Target/output labels.

Basic Viva Questions & Answers

Q1. What is the purpose of this program?

Answer: To divide a dataset into training and testing data.

Q2. What is a dataset?

Answer: A collection of related data stored in rows and columns.

Q3. How many datasets are created in this program?

Answer: Two datasets:

1. Training Dataset
2. Testing Dataset

Q4. Why do we split the dataset?

Answer: To train the model and test its performance separately.

Q5. What is machine learning?

Answer: Machine learning is a technique that enables computers to learn from data and make predictions.

Q6. What is training data?

Answer: Data used to teach or train the model.

Q7. What is testing data?

Answer: Data used to check the accuracy of the model.

Q8. What is the full form of NumPy?

Answer: Numerical Python.

Q9. Why do we use NumPy?

Answer: To create and manage arrays efficiently.

Q10. What is an array?

Answer: A collection of similar data elements stored together.

Questions Based on Code

Q11. What does import numpy as np mean?

Answer: It imports the NumPy library and assigns it the alias name np.

Q12. What does np.array() do?

Answer: It creates an array.

Q13. What is stored in X?

Answer: Input features (Age and Salary).

Q14. What is stored in y?

Answer: Output labels or target values.

Q15. What are features?

Answer: Input variables used to make predictions.

Q16. What is the target variable?

Answer: The output value that we want to predict.

Q17. How many features are present in X?

Answer: Two features:

- Age
- Salary

Q18. How many records are present in the dataset?

Answer: Six records.

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train_test_split Questions

Q19. Which function is used for dataset splitting?

Answer: train_test_split()

Q20. Which library provides train_test_split()?

Answer: Scikit-Learn (sklearn).

Q21. What does test_size=0.3 mean?

Answer: 30% of data is used for testing.

Q22. What does random_state=42 mean?

Answer: It ensures the same random output every time.

Q23. What happens if random_state is removed?

Answer: Different records may be selected each time the program runs.

Q24. Why is random splitting used?

Answer: To avoid bias in data selection.

Q25. How much training data is used here?

Answer: 70%.

Q26. How much testing data is used here?

Answer: 30%.

Logical Questions

Q27. Can we train a model without training data?

Answer: No, because the model learns from training data.

Q28. Can we evaluate a model without testing data?

Answer: No, testing data is required to check performance.

Q29. Which dataset should be larger: training or testing?

Answer: Training dataset should be larger.

Q30. Why should training data be larger?

Answer: More data helps the model learn better.

Q31. What is the benefit of testing data?

Answer: It helps measure model accuracy on unseen data.

Q32. What is overfitting?

Answer: When a model performs well on training data but poorly on new data.

Q33. How does train-test split help?

Answer: It helps detect overfitting and evaluate performance.

Output-Based Questions

Q34. What does X_train contain?

Answer: Training feature values.

Q35. What does X_test contain?

Answer: Testing feature values.

Q36. What does y_train contain?

Answer: Training labels.

Q37. What does y_test contain?

Answer: Testing labels.

Q38. Why do we print X_train?

Answer: To see the training dataset.

Q39. Why do we print X_test?

Answer: To see the testing dataset.

Q40. Why do we print y_train and y_test?

Answer: To verify the corresponding output labels.

One-Line Answers (Most Important)

What is NumPy?
A Python library for numerical computations.

What is Scikit-Learn?
A machine learning library in Python.

What is train_test_split()?
A function used to divide data into training and testing sets.

What is X?
Input features.

What is y?
Target/output labels.

What is random_state?
A parameter that makes results reproducible.

What is test_size?
The proportion of data used for testing.

Why split data?
To train and evaluate the model separately.